🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.9

Determine Distance on the Coordinate Plane

Lesson 7-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the distance between two points on the coordinate plane using absolute value.

Language Objective: I can explain my work using the words distance, absolute value, horizontal distance, and vertical distance.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Distance | Absolute value | Horizontal distance | Vertical distance | Coordinate plane | Integer

👆 *Tap any word to see its meaning and a picture.*

1. How far apart two points are on the coordinate plane is the ____________.

2. The distance of a number from zero, used to find distance between points, is its ____________.

3. The distance between two points along the x-direction is the ____________.

4. The distance between two points along the y-direction is the ____________.

5. The grid made by the x-axis and y-axis is the ____________.

6. A whole number or its opposite is an ______.

Write About the Math

How does the drone use coordinate distance to calculate the total flight path?

¿Cómo usa el dron la distancia en coordenadas para calcular la ruta de vuelo total?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Distance** — *Distancia*
- ☐ **Absolute value** — *Valor absoluto*
- ☐ **Horizontal distance** — *Distancia horizontal*
- ☐ **Vertical distance** — *Distancia vertical*
- ☐ **Coordinate plane** — *Plano cartesiano*

Model: *I can find the distance by looking at the coordinates that ____.* — Student finds the distance by subtracting the x-coordinates (7 - 2 = 5) since the points share a horizontal line.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The horizontal distance from (-4, 0) to (3, 0) is ____ units because ____. The vertical distance from (3, 0) to (3, -5) is ____ units because ____. The total flight path is ____ units.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Distance
2. **Notes 2:** Absolute value
3. **Notes 3:** Horizontal distance
4. **Notes 4:** Vertical distance
5. **Notes 5:** Coordinate plane
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Distance on the Coordinate Plane is subtracting the coordinates as if both points were on the same side of zero, instead of adding their absolute values when the points are on opposite sides. For example, to find the distance between (-6, 3) and (2, 3), a student might subtract 6 − 2 = 4, but since -6 and 2 are on opposite sides of zero, the absolute values must be added instead: |-6| + |2| = 6 + 2 = 8. The correct distance is 8 units, not 4. Before you submit, ask: 'Are these two points on opposite sides of zero — and if so, did I add the absolute values instead of subtracting?'*
8. **Try It 1:** B. 7 blocks — *Same y = 5, so the segment is horizontal. The points are on opposite sides of zero, so add the absolute values: |-4| + |3| = 4 + 3 = 7 blocks.*
9. **Try It 2:** B. 5 blocks — *Same x = -2, so the segment is vertical. The points are on opposite sides of zero: |1| + |-4| = 1 + 4 = 5 blocks.*